FRONTEND DEVELOPMENT • LEARNING PATH

REACT COMPLETE ROADMAP

A structured path from JavaScript prerequisites to React projects, advanced tools, optimization, and next steps.

08 CORE PHASES	**03** PROJECT LEVELS	**01** CLEAR PATH

DARK • CLEAN • ORANGE ACCENTS

Use this roadmap as a checklist. Move from fundamentals to rendering, APIs, routing, hooks, advanced ecosystem tools, and real projects.

ROADMAP CONTENT

Prerequisites / React Basics / Rendering / APIs / Routing / Hooks / Advanced Powers / Projects / Optimization / Next Steps

CONTENTS

A quick index of the roadmap sections.

01	Prerequisite	
02	React Basics	
03	Rendering	
04	APIs	
05	Routing	
06	Hooks	
07	Advance Powers	
08	Projects	
09	Optimizations	
10	After React	
11	Optional — Git & GitHub	

Numbering guide: main topics use section-based numbers (for example, 1.1), and nested items use the next level (for example, 1.1.1).

PHASE 01

Prerequisite

1.1 Basics

1.1.1 Variables, operators, etc.

1.2 Functions

1.2.1 Core concepts and practice

1.3 Array, Object, String

1.3.1 Core concepts and practice

1.4 DOM Manipulation

1.4.1 Core concepts and practice

1.5 Promises & APIs

1.5.1 Core concepts and practice

1.6 Destructuring

1.6.1 Core concepts and practice

1.7 Modules

1.7.1 Core concepts and practice

PHASE 02

React Basics

2.1 CLI Tool

2.1.1 Vite
2.1.2 Create React App

2.2 Folder Structure

2.2.1 Core concepts and practice

2.3 Components

2.3.1 JSX
2.3.2 Props
2.3.3 States (useState)
2.3.4 Data Flow

2.4 Hooks

2.4.1 Core concepts and practice

2.5 Styling

2.5.1 CSS
2.5.2 Tailwind CSS

PHASE 03

Rendering

3.1 Event Handling

3.1.1 Core concepts and practice

3.2 Life cycle

3.2.1 useEffect

3.3 Conditional Rendering

3.3.1 Core concepts and practice

3.4 List Rendering

3.4.1 Normal Array
3.4.2 Array of Objects
3.4.3 Array of Array

3.5 Refs

3.5.1 useRef

3.6 Higher Order Components

3.6.1 Core concepts and practice

PHASE 04

APIs

4.1 Fetch

4.1.1 Core concepts and practice

4.2 Axios

4.2.1 Core concepts and practice

4.3 React Query

4.3.1 Core concepts and practice

PHASE 05

Routing

5.1 React Router

5.1.1 Core concepts and practice

5.2 TanStack Router

5.2.1 Core concepts and practice

PHASE 06

Hooks

6.1 useState

6.1.1 Core concepts and practice

6.2 useEffect

6.2.1 Core concepts and practice

6.3 useRef

6.3.1 Core concepts and practice

6.4 useLocation

6.4.1 Core concepts and practice

6.5 useCallback

6.5.1 Core concepts and practice

6.6 useMemo

6.6.1 Core concepts and practice

PHASE 07

Advance Powers

7.1 STATE MANAGEMENT

7.1.1 Context API + Reducer
7.1.2 Zustand
7.1.3 Redux
7.1.4 Redux Toolkit

7.2 STYLING

7.2.1 CSS
7.2.2 Tailwind CSS
7.2.3 Daisy UI
7.2.4 Styled Components
7.2.5 ShadCN
7.2.6 Material UI
7.2.7 Ant Design
7.2.8 Redux UI

7.3 ANIMATIONS

7.3.1 Gsap
7.3.2 React Spring
7.3.3 Framer Motion

7.4 VALIDATION

7.4.1 React Hook Form
7.4.2 Zod
7.4.3 Formik
7.4.4 Yup

7.5 TESTING

7.5.1 Jest
7.5.2 Vitest
7.5.3 React Testing Library

PHASE 08

Projects

8.1 BEGINNER

- 8.1.1 To-Do List
- 8.1.2 Weather App
- 8.1.3 Calculator
- 8.1.4 Mini E-commerce Store
- 8.1.5 Quiz App
- 8.1.6 Expense Tracker
- 8.1.7 News App

8.2 INTERMEDIATE

- 8.2.1 Basic E-commerce Store
- 8.2.2 Blog App
- 8.2.3 Chat App
- 8.2.4 Movie App
- 8.2.5 Portfolio Website

8.3 ADVANCE

- 8.3.1 Advance E-commerce Store
- 8.3.2 LMS
- 8.3.3 Yt Clone
- 8.3.4 Task Management App
- 8.3.5 Social Media Clone

OPTIMIZATIONS

Performance & SEO

9.1 Performance Optimization

- 9.1.1 useMemo
- 9.1.2 code splitting
- 9.1.3 virtualization

9.2 Bundle Size Optimization

- 9.2.1 Core concepts and practice

9.3 Cost Optimizations

- 9.3.1 Reduce api calls
- 9.3.2 Lazy loads (load only what's needed)

9.4 SEO

- 9.4.1 Core concepts and practice

AFTER REACT

Next Steps

10.1 Next JS

- 10.1.1 Core concepts and practice

10.2 React Native (Android)

- 10.2.1 Core concepts and practice

10.3 Recommended

10.3.1 Backend

OPTIONAL

Version Control

11.1 VCS

11.1.1 Git & GitHub